

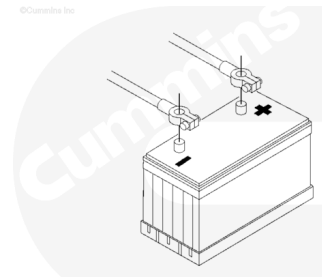
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries
- Remove the rocker lever cover and gasket. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html>).



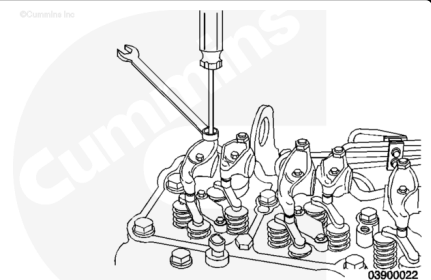
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Remove

B4.5 RGT Engines

Loosen the adjusting screw locknuts.

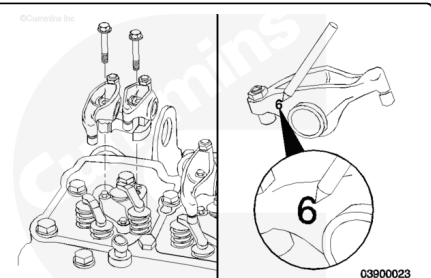
Loosen the adjusting screws until they stop.



Remove the capscrews from the rocker lever pedestals.

Remove and mark the pedestals and rocker lever assemblies one at a time as to their location and position.

Remove and mark the crossheads one at a time as to their location and position. The crossheads and rocker assembly **must** be installed in their original location and position when

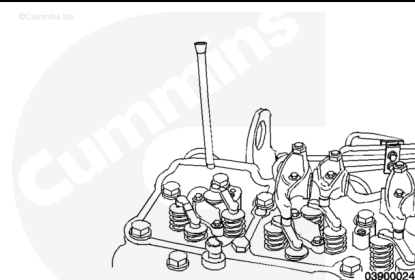


reused.

Mark the push rods to identify their location.

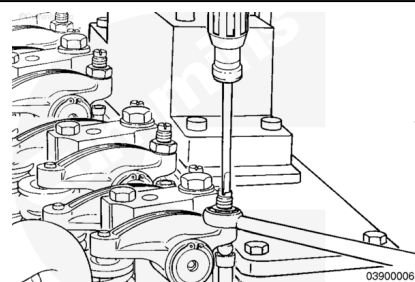
Note : The push rods must be installed in their original location and position.

Remove the push rods. Refer to Procedure 004-014 (</qs3/pubsys2/xml/en/procedures/40/40-004-014-tr.html>).

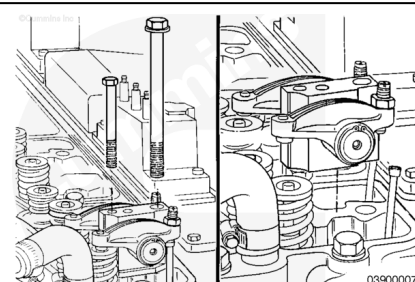


B3.9, B5.9, and B4.5 Engines

Loosen the adjusting screw locknuts. Loosen the adjusting screws until they stop.

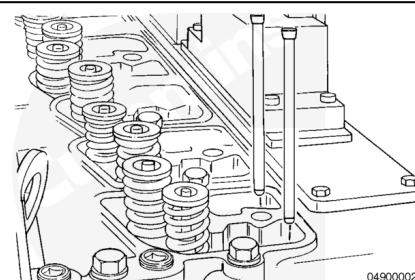


Remove the capscrews from the rocker lever pedestals.
Remove the pedestals and rocker lever assemblies.



Note : Mark the push rods to identify their location.

Remove the push rods.

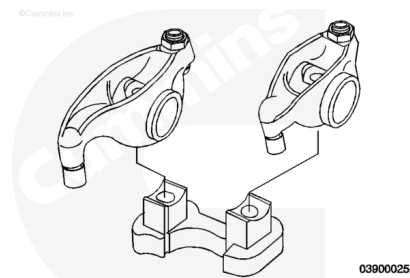


Disassemble

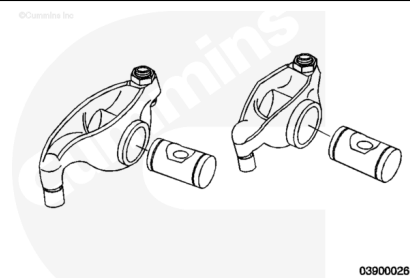
B4.5 RGT Engines

Note : All rocker lever components **must** be installed in their original location and position.

Remove the rocker levers from the pedestal.



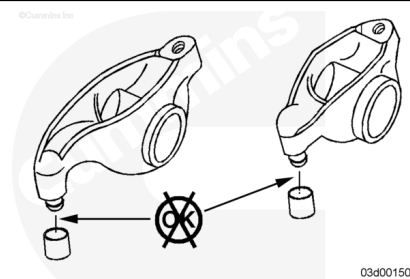
Remove the rocker lever shafts from the rocker levers.



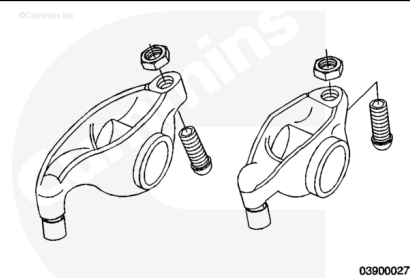
The socket at the tip of the rocker lever should **not** be removed.

This part is **not** serviceable.

If damage to the socket is found the rocker lever **must** be replaced.



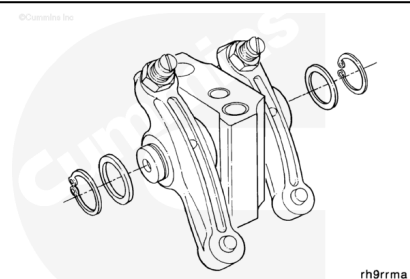
Remove the locknut and adjusting screw.



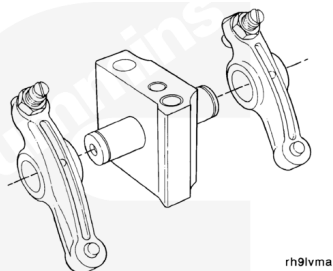
B3.9, B5.9, and B4.5 Engines

If the rocker lever and push tubes are to be inspected for reuse, follow these steps.

Remove the retaining rings and thrust washers.

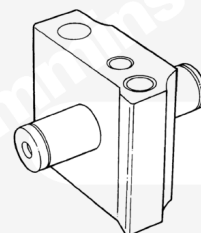


Remove the rocker levers.



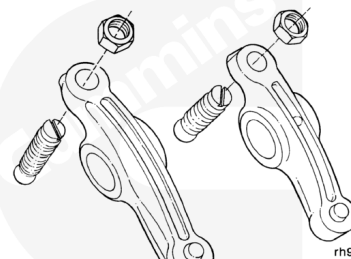
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Note : Do not disassemble the rocker lever shaft and pedestal. The pedestal and shaft must be replaced as an assembly.



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Remove the locknut and adjusting screw.

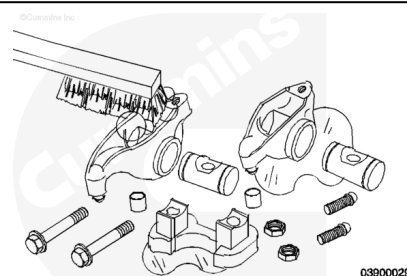


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Clean and Inspect for Reuse

B4.5 RGT Engines

Clean all parts in a strong solution of laundry detergent and hot water.



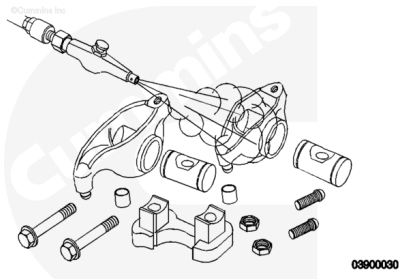
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⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use compressed air to dry the parts after rinsing in clean, hot water.

Note : The pedestals are made from powdered metal and will appear wet even after they have been cleaned and dried.

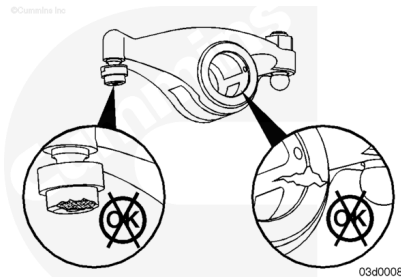


Inspect for cracks and excessive wear in the bore.

The socket should move freely on the rocker lever and the plastic socket retainer should be in place and not cracked.

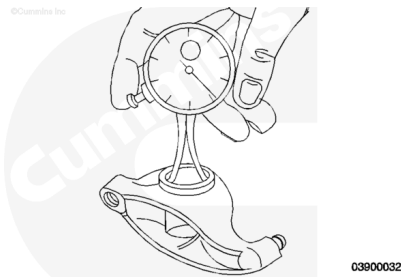
Inspect for wear on the nose of the socket.

Note : The socket at the tip of the rocker lever should not be removed. This part is not serviceable. If damage to the socket is found the rocker lever must be replaced.



Measure the rocker lever bore.

mm		in
22.027	MAX	0.867

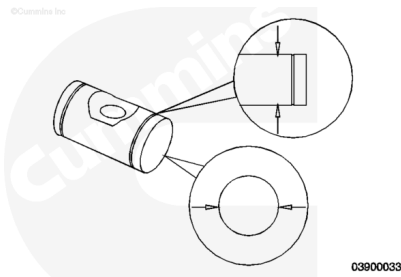


Inspect the rocker lever pedestal and rocker lever shaft.

It is **not** uncommon for the rocker lever shaft to have a polished appearance. Inspect for the following to determine rocker lever shaft reusability.

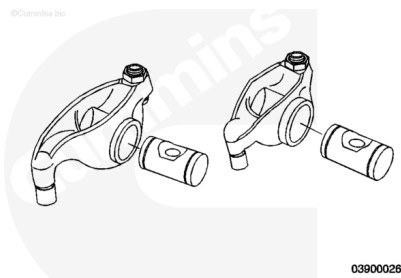
Measure the rocker lever shaft diameter.

mm		in
21.965	MIN	0.865



Lubricate the rocker lever shaft with clean 15W-40 engine lubricating oil.

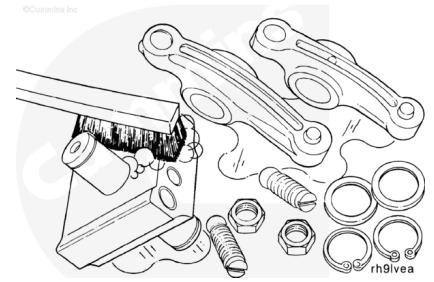
Install the rocker lever shaft into the rocker lever and rotate the rocker lever 180 degrees about the rocker lever shaft. The rocker lever should rotate freely without any binding.



If the rocker lever does **not** rotate freely, replace the rocker lever shaft and rocker lever.

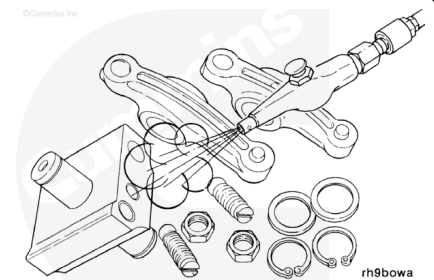
B3.9, B5.9, and B4.5 Engines

Clean all parts in a strong solution of detergent in hot water.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam or high-pressure water can cause serious personal injury.



⚠ WARNING ⚠

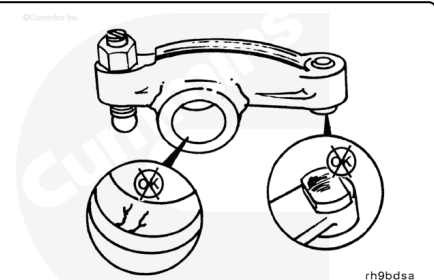
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Rinse parts with clean, hot water.

Dry with compressed air.

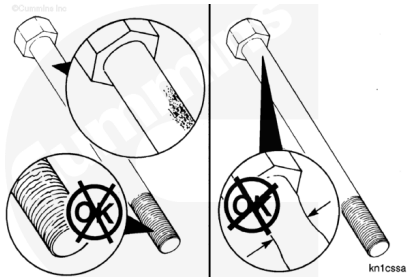
Note : The pedestals are made from powdered metal and will continue to show wetness after they have been cleaned and dried.

Inspect for cracks and excessive wear in the bore and the contact surface of the valve stem.



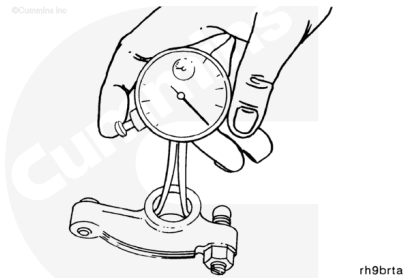
The longer rocker lever capscrew also joins the cylinder head to the cylinder block.

Inspect the cylinder head capscrew for damaged threads, corroded surfaces, reduced diameter and increased length (due to capscrew stretching). Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html).



Measure the rocker lever bore.

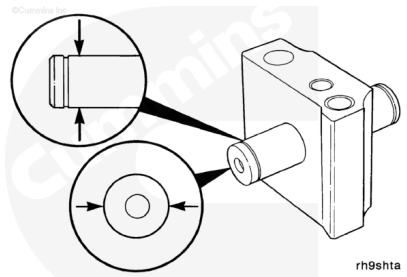
mm		in
19.00	MIN	0.748



Inspect the pedestal and shaft.

Measure the shaft diameter.

mm		in
18.98	MAX	0.747

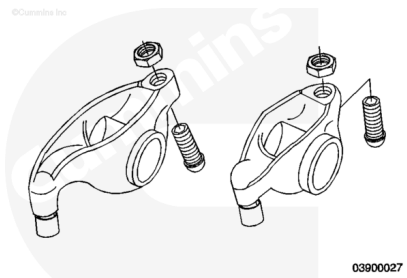


Assemble

B4.5 RGT Engines

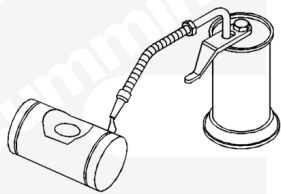
Note : All rocker lever components must be installed in their original location and position.

Install the adjusting screw until it stops and then install the locknut.



Lubricate the rocker lever shafts with clean 15W-40 engine lubricating oil.

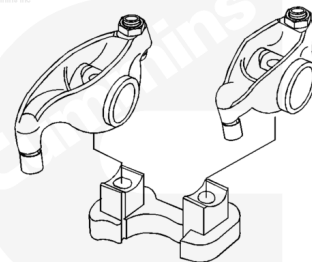
Install the rocker lever shafts into the rocker levers.



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Position the rocker levers on the rocker pedestal.

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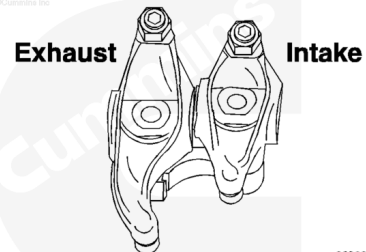


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⚠ CAUTION ⚠

Be sure to assemble the intake and exhaust rocker levers in the correct location. Failure to do so will result in engine damage.

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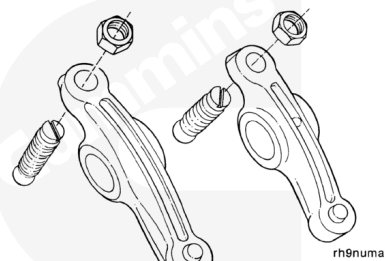


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B3.9, B5.9, and B4.5 Engines

Install the adjusting screw and locknut.

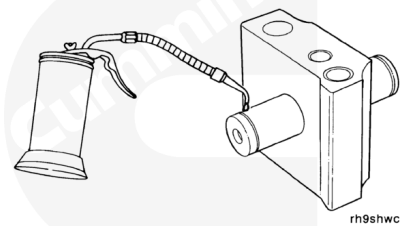
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Lubricate the shaft with clean lubricating engine oil.

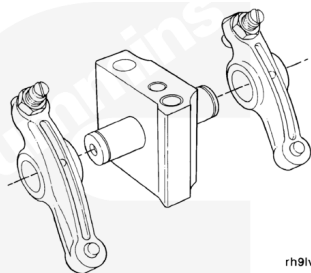
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Position the levers on the rocker shaft.

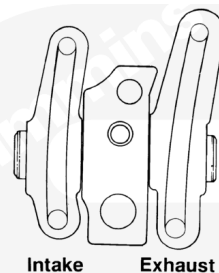
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rh9lvma

Note : Be sure to assemble the intake and exhaust rocker levers in the correct location.

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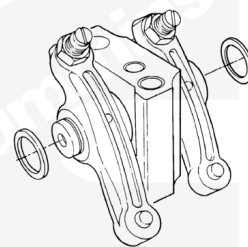
Intake

Exhaust

rh9lvba

Install the thrust washers.

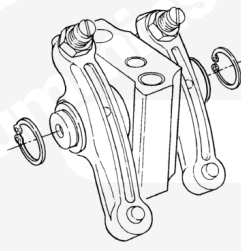
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Install the snap rings.

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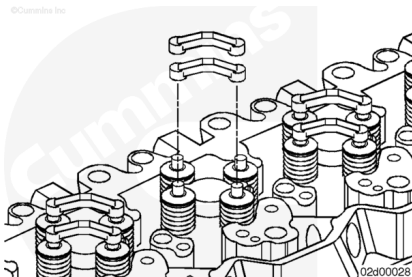


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Install

B4.5 RGT Engines

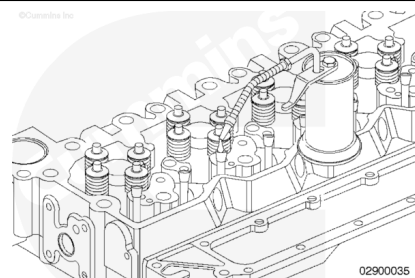
Install the crossheads in their original location and position.



Install the push rods as marked during removal.

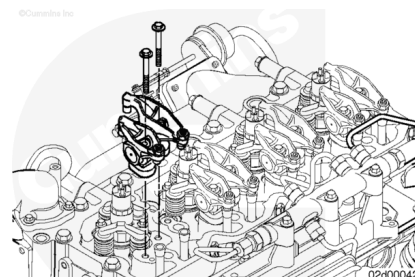
Install the push rods into the sockets of the valve tappets.

Lubricate the push rod sockets with clean 15W-40 lubricating engine oil.



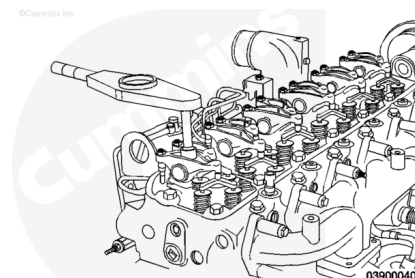
Install the rocker lever assemblies and pedestals in their original position.

Install the pedestal mounting capscrews.



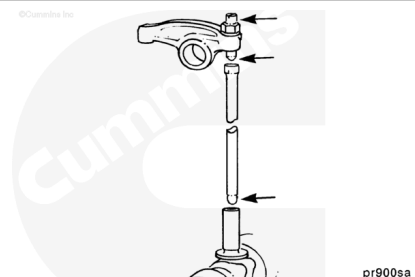
Tighten the pedestal mounting capscrews.

Torque Value: 36 N·m [27 ft-lb]

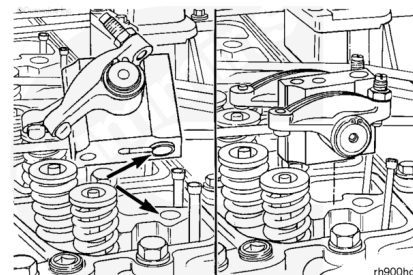


⚠ CAUTION ⚠

Rotate the adjusting screws until the adjusting screw contacts the socket of the push rod. This will ensure the push rod stays in contact with the tappet and adjusting screw when the engine is rotated to set valve lash. If not completed, the push rods may be bent or damaged.

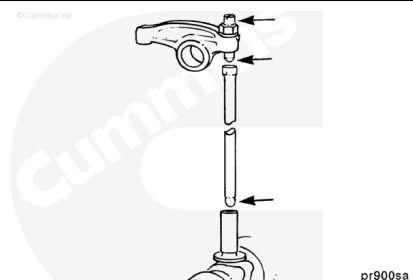


Make sure the dowel rings in the pedestals are installed into the dowel bores.



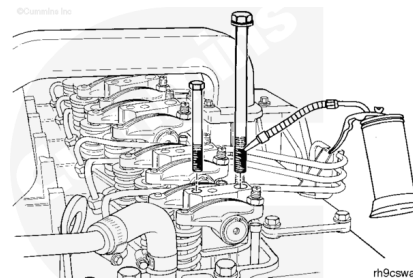
⚠ CAUTION ⚠

Rotate the adjusting screws until the adjusting screw contacts the socket of the push rod. This will ensure the push rod stays in contact with the tappet and adjusting screw when the engine is rotated to set valve lash. If not completed, the push rods may be bent or damaged.

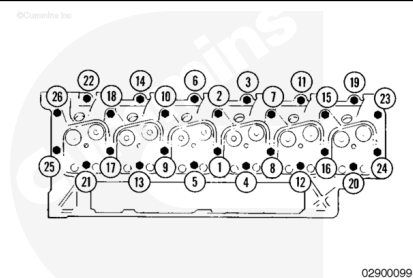


Use clean lubricating engine oil to lubricate the threads and under the heads of the capscrews.

Install the pedestal and cylinder capscrews into the pedestals.

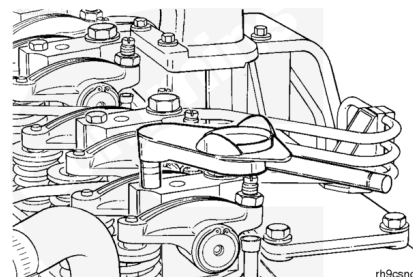


Tighten all the cylinder head capscrews following proper torque order. Refer to Procedure 002-004 ([/qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html](http://qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html)).



Tighten the 8-mm pedestal capscrews.

Torque Value: 24 n•m [18 ft-lb]



Finishing Steps

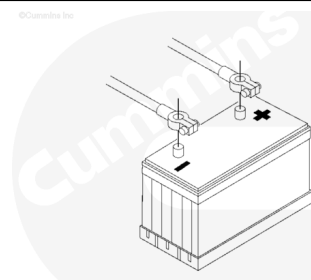
- Set valve lash for intake and exhaust valves. Refer to Procedure 003-004 (</qs3/pubsys2/xml/en/procedures/40/40-003-004-tr.html>)
- Install the rocker lever cover gasket and rocker lever cover. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html>).



ck800wa

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Connect the batteries
- Run the engine and check for leaks.